

Project Summary

System: Web-based voting platform with 3 user roles (Voter, Election Officer, System Admin).

Name: Smart Choice Online Voting System. (An online voting app for university students)

Core Tech:

- Backend: PHP
- Frontend: HTML/CSS/JavaScript
- Database: Supabase (PostgreSQL) or free alternative
- Media: Cloudinary or free alternative
- Hosting: Vercel or free alternative

Key Features: Voter registration, secure voting with OTP, real-time dashboards, PDF results, and admin controls.

Core Requirements

1. Home Page

- **Design:** Light blue theme :
 - Voter Registration
 - User Login
 - System Admin
- Responsive on all devices

2. User Registration (Voters Only)

- **Fields:**
First Name, Last Name, Email, Department, Student_ID, Profile Image, Gender (dropdown), Password (min 8 chars), Re-enter Password.
- **Validation:**
 - Passwords must match. Should store in an encrypted manner
 - Prevent duplicate Email/Employee Number (voters only)
 - "Register" button enabled only when all fields filled

- **Success:** Redirect to registration page with confirmation message

3. Login System

- **Login Types:**
 - Voter Login
 - Election Officer Login (for admin-approved users)
- **Password Reset:** Forgot password flow via email
- **Validation:** Show errors for incorrect credentials

4. Voter Dashboard

- **Features:**
 - **Profile:** View/update Department and Profile Image
 - **Self-Enrollment:** Request to become Candidate/Election Officer
 - **Delete Account:** Removes voter without affecting election history
 - **Icons:**
 - *On-Going Elections:* Vote in active elections
 - *Results:* View/download past election PDF reports
- **Voting Process:**
 - OTP sent to email for vote confirmation (valid 15 mins)
 - Prevent duplicate votes
 - Encrypt votes

5. Election Officer Dashboard

- **Create Elections:**
 - Set name, date, positions, voting period
 - Assign eligible voters/candidates (1 position per candidate), departments
 - Add "None of the Above" option per position
- **Monitor Elections:**
 - Real-time voting stats (no individual vote data)

- Should not be able to see who voted to whom
- Auto-publish results as PDF after voting ends

6. System Admin Dashboard

- **Auth:** Predefined superuser credentials
- **Functions:**
 - Approve/remove Election Officer requests
 - Delete voter accounts
 - View/download audit logs
 - Backup/recovery
 - Custom reports (PDF/Excel)

Technical Specifications

1. **Architecture:** MVC pattern
2. **Performance:**
 - Caching for UI speed
 - Optimized database queries
3. **Security:**
 - Input validation on all forms
 - Passwords should be encrypted
 - Error logging (frontend + backend)
4. **Design:**
 - Consistent light blue theme
 - Simple UI

Deliverables

1. Complete PHP web application with:
 - Database schema for Supabase
2. Documentation:
 - Setup instructions to run the app, how to setup Eg: Vercel/Supabase/Cloudinary
 - Code comments explaining key logic

Sample LOGO



Smart Choice: Online
Voting System